ENTERPRISE SOC ARCHITECTURE & TECHNICAL MANUAL

SentinelAI X — MCP Security Copilot

Enterprise Autonomous SOC Platform with Real Wazuh SIEM Telemetry, 16 Model Context Protocol (MCP) Tools, and Machine Learning Anomaly Detection

Executive Summary: SentinelAI X is an autonomous next-generation Security Operations Center (SOC) platform engineered to solve alert fatigue, accelerate incident response, and bridge the forensic intelligence gap. Powered by a high-speed React 18 frontend and Python FastAPI backend, the system establishes a live telemetry bridge to **Wazuh SIEM v4.14.7**, orchestrates **16 Model Context Protocol (MCP)** threat intelligence tools (featuring 6 zero-key public intelligence feeds), and deploys an enterprise **scikit-learn Machine Learning Anomaly Engine** (IsolationForest + RandomForest + UEBA drift analysis). This document details every feature, architectural component, and operational workflow.

Project Repository:	https://github.com/MatheshT/mcp-security-copilot
Author / Engineer:	Mathesh T (Lead Security Developer)
Platform Version:	v2.0.0 (Enterprise Cyber Black & Matrix Green Edition)
Backend Engine:	FastAPI 0.104+, Uvicorn, Python 3.11, SQLite/PostgreSQL
Frontend UI:	React 18, Vite 5, Custom Matrix Green Design System
Integrated SIEM:	Wazuh v4.14.7 OVA via Autonomous Headless VirtualBox
Connected MCP Tools:	16 Total Tools (MalwareBazaar, CISA KEV, IP-API, CIRCL CVE, etc.)
Release Date:	October 2026

1. System Architecture & End-to-End Workflow

SentinelAI X is engineered as a decoupled, micro-service architecture optimized for speed, fault tolerance, and modular expansion. The entire pipeline operates seamlessly from endpoint log generation down to active network containment.

Layer	Component	Technology	Role & Capabilities
Presentation	SOC Frontend UI	React 18, Vite 5, Custom CSS	Real-time HUD header, Cyber Black & Matrix Green theme, live triage console, AI copilot terminal, audio cues.
API & Routing	Core Security API	FastAPI, Uvicorn, Pydantic	Asynchronous REST endpoints, JWT authentication, non-blocking telemetry caching, rate limiting.
Intelligence Protocol	MCP Client & Tool Hub	Model Context Protocol (MCP)	Standardized tool orchestration interface connecting 16 discrete intelligence providers into prompt context.
SIEM & Telemetry	Fleet Monitoring	Wazuh v4.14.7 OVA, VirtualBox	Live agent correlation, FIM (syscheck), Windows Defender events, PAM auth, sub-second JSON log tailing.
Machine Learning	ML Threat Engine	scikit-learn, NumPy, Joblib	IsolationForest anomaly scoring (0-100), RandomForest severity prediction, UEBA baseline drift calculation.
Orchestration	Active Defense (SOAR)	Windows Netsh, Subnet Guard	Zero-click firewall blackholing, infected host network quarantine, rollback capability.

End-to-End Incident Lifecycle Execution Flow:

1. **Telemetry Ingestion:** Endpoint events (Windows Security Event ID 4624/4625, Syscheck FIM, Sysmon) are streamed to Wazuh Manager inside the VirtualBox VM.
2. **Sub-Second Log Parsing:** The backend engine tails /var/ossec/logs/alerts/alerts.json and parses rule IDs, rule levels, agent metadata, and full event payloads.
3. **In-Memory & Disk Persistence:** Real alerts are indexed into _alerts_cache and synchronized to wazuh_alerts_cache.json, enabling instant 0ms responses on startup.
4. **Machine Learning Inference:** Events are vector-analyzed via IsolationForest (for zero-day anomalies) and RandomForest (predicting Severity and False Positive likelihood).
5. **Multi-Tool MCP Enrichment:** Extracted IOCs (IPs, Hashes, CVEs) are dispatched simultaneously to free tools (MalwareBazaar, CISA KEV, IP-API, CIRCL).
6. **Analyst Triage & Remediation:** The analyst reviews live alerts or interacts with SentinelAI Copilot. Single-click containment rules trigger firewall drops or network isolation.

2. Complete Feature Breakdown

Feature 1: Real-Time SOC Command Center & HUD Header

The primary landing dashboard gives security operators an immediate executive and operational pulse of the organization. Key capabilities include:

- **Telemetry Ticker Stream:** Top HUD stream displays real-time ticker tags including active defense shield status, live alert counters, and UTC timestamp synchronization.
- **Global Threat Radar:** Visual radar displaying monitored endpoints, connection latency, and agent status.
- **Keyboard Shortcuts Navigation:** Fast one-key navigation across consoles (1: Command Center, 2: AI Copilot, 3: ML Engine, 4: Real Wazuh Alerts, 5: Threat Intel, 6: Swarm Hub, 7: Reports).
- **Cyber Audio Telemetry:** Sub-audible click, warp, alert, and success frequencies provide sensory situational awareness.

Feature 2: Real Wazuh SIEM v4.14.7 Integration & Live Triage

Unlike platforms that present synthetic or mock alert data, SentinelAI X is connected directly to a live Wazuh SIEM instance running in VirtualBox. It delivers production-grade SIEM capabilities:

- **Native alerts.json Parser:** Sub-second parsing of real alerts including Auditd SELinux checks, PAM authentication sessions, and Registry modifications from monitored Windows/Linux endpoints.
- **Zero-Lag Startup Architecture:** Utilizes pre-populated memory caching and non-blocking VM status verification, completely resolving cold-boot latency and timeout errors.
- **Triage & Incident Management:** Filter by severity (Critical, High, Medium, Low), search across full logs, update status (Open, Investigating, Resolved), and persist analyst investigation notes.
- **Client-Side Persistence:** React localStorage caching guarantees the alerts table renders in 0 milliseconds the moment the user opens the platform.

Feature 3: SentinelAI Security Copilot (Hero Chatbot)

The Security Copilot serves as an AI force multiplier for SOC Tier-1 and Tier-2 analysts. It can autonomously investigate alerts, explain technical rule logs, and guide threat containment:

- **Natural Language Security Dialogue:** Query any security concept, incident log, or system status in plain English or Tamil/Tanglish (e.g., *'Wazuh alert check pannu ethana alerts iruku'*).
- **Autonomous Intent & IOC Extraction:** Automatically detects IP addresses, CVE identifiers, and file hashes in analyst queries and routes them to the correct MCP tools.
- **Clean Cyber Terminal Interface:** Clutter-free terminal interface with quick tactical prompt chips, copyable forensic code blocks, and real-time execution animation.
- **Investigation Markdown Export:** One-click export of complete multi-turn forensic investigations into professional Markdown (.md) reports for compliance and auditing.

Feature 4: Enterprise Machine Learning Anomaly & Threat Engine

SentinelAI X incorporates advanced scikit-learn models running directly in Python to detect subtle, multi-vector attacks that bypass static SIEM correlation rules:

ML Model / Pipeline	Algorithm / Framework	Input Features	Security Deliverable
Anomaly Detector	IsolationForest (Unsupervised)	Event frequency, rule level, off-hours execution, LOLBin flags	Calculates Anomaly Risk Score (0–100) identifying zero-day and anomalous behavior.
Severity Predictor	RandomForest Classifier (Supervised)	Failed login ratio, attack phase, user privilege level, service context	Predicts Critical/High/Medium/Low severity and outputs False-Positive probability.
UEBA Drift Engine	Behavioral Gaussian Drift	Login timestamp variance, subnet hopping, process divergence	User & Entity Behavior Analytics score flagging compromised credentials.

Feature 5: 16 Model Context Protocol (MCP) Tools (6 Free SOC Feeds)

The Model Context Protocol (MCP) enables standardized, bidirectional communication between AI models and forensic tools. SentinelAI X integrates 16 tools, highlighted by **6 completely free, zero-key SOC feeds**:

Tool Name	Category	Zero-Key Free	Capabilities & Value
abuse.ch MalwareBazaar	Threat Intel	**YES (100% Free)**	Query public malware samples, SHA256/MD5 hashes, and signatures without API keys.
abuse.ch URLhaus	Payload Tracking	**YES (100% Free)**	Real-time database of active malware distribution sites and C2 infrastructure URLs.
CISA KEV Catalog	Vulnerability Intel	**YES (100% Free)**	Official US DHS catalog of weaponized CVEs actively exploited by threat actors.
IP-API Geolocation	Network Intel	**YES (100% Free)**	Free IP ASN, ISP, country, city, reverse DNS PTR, and hosting proxy detection.
CIRCL / OSV CVE Search	Vulnerability Search	**YES (100% Free)**	Free lookup of CVSS v3.1 base metrics, vector strings, and official vendor advisories.
MITRE ATT&CK; Matrix	Adversary TTPs	**YES (100% Free)**	Tactical mapping of techniques (e.g. T1059, T1110, T1486) and defensive mitigations.

ML Threat Engine	Machine Learning	**YES (Local Model)**	Direct prompt context injection of IsolationForest scores and UEBA drift telemetry.
Wazuh SIEM	SIEM & Fleet	**YES (Local VM)**	Retrieves live fleet telemetry, security events, and agent keepalive metrics.
Automated Containment	SOAR Defense	**YES (Local SOAR)**	Applies gateway firewall drop rules and host network quarantine.
WindowsLogs / Defender	Endpoint Auditing	**YES (Local OS)**	Ingests Windows Event IDs 4624/4625 and real-time Microsoft Defender signals.
VirusTotal / AbuseIPDB	Commercial Intel	API Key Supported	Multi-engine antivirus analysis (70+ scanners) and crowdsourced IP reputation scores.

Feature 6: Automated Threat Containment & Active Defense

SentinelAI X empowers security teams to transition from passive observation to automated active containment. Containment actions can be executed directly from the AI chat copilot or alert triage cards:

- **Firewall IP Blackholing:** Automated gateway-level drop rule creation (e.g. blocking malicious C2 IP 185.220.101.45) via Windows Netsh or iptables.
- **Endpoint Subnet Isolation:** Isolates compromised workstations from internal network subnets while maintaining telemetry communication to the Wazuh SIEM.
- **Policy Rollback & Audit Trail:** Every containment policy execution is timestamped and recorded in the audit log with instant one-click rollback capability.

Feature 7: Visual Cyber Black & Matrix Green Design System

The user interface has been custom-styled with an authentic cyber terminal aesthetic engineered for high readability, low operator eye fatigue during extended SOC shifts, and high visual appeal:

- **Obsidian Black Foundations:** Deep pitch-black canvases (#000000, #020603) replacing generic slate gray backgrounds.
- **Matrix Emerald Green Accents:** Neon high-contrast accents (#10b981, #00ff66, #34d399) highlighting active routes, buttons, borders, and radar sweeps.
- **Modern Typography:** Crisp JetBrains Mono and Inter typography with high-contrast text rendering.

3. Operations, Startup & Verification Guide

Zero-Click Automated Startup:

The project includes intelligent launcher scripts (run.bat and start.bat) that execute the complete bootstrap process automatically:

1. Verifies Python 3.10+ and Node.js v18+ environment binaries.
2. Terminates any stale processes holding network ports **8000** or **5173**.
3. **Autonomous VM Boot:** Detects VirtualBox and boots Wazuh v4.14.7 OVA in headless background mode silently.
4. **Starts FastAPI SOC Backend:** Boots Uvicorn on http://127.0.0.1:8000 with pre-loaded alert caches.
5. **Starts React Vite Frontend:** Boots development server on http://127.0.0.1:5173.
6. Automatically opens your default web browser to the live platform.

System Health & API Verification Matrix:

Target Endpoint	HTTP Method	Expected Response	System Role
/api/health	GET	{"status": "operational"}	Verifies backend services, ML engine bootstrap, and database connection.
/api/detection/wazuh/alerts	GET	{"total_alerts": 100+, ...}	Streams real-time parsed Wazuh SIEM security events and rule telemetry.
/api/investigate	POST	{"ai_answer": ..., "required_tools": [...]}	AI Copilot multi-tool investigation and IOC correlation pipeline.
/api/ml/detect-anomaly	POST	{"anomaly_score": 57.6, ...}	Executes IsolationForest anomaly inference on submitted event telemetry.
/api/soar/actions	GET / POST	{"status": "contained"}	Executes and logs active defense firewall rules and network isolation.

Architectural Certification: *SentinelAI X v2.0 represents a production-grade integration of modern Autonomous AI Agents, Model Context Protocol standards, Enterprise SIEM correlation, and Applied Machine Learning. All modules are tested, operational, and synchronized with the official GitHub repository.*